

RUOQIANG, China

WMO: 517770

Lat: 39.0239N

Lon: 88.1837E

Elev: 2917

StdP: 13.21

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.3	6.5	-9.3	3.7	15.7	-6.2	4.4	18.6	14.7	25.1	12.5	22.3	3.2	250	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.2	100.9	66.7	98.5	65.8	96.0	65.1	71.4	91.0	69.4	89.5	67.9	89.0	5.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	105.6	79.9	63.2	96.7	78.2	61.0	89.3	77.4	37.3	90.6	35.4	89.4	34.0	88.8	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	15.4	12.7	-0.1	106.2	3.7	2.1	-2.7	107.8	-4.9	109.0	-6.9	110.2	-9.6	111.8		
			-1.0	77.3	3.4	2.7	-3.4	79.2	-5.4	80.8	-7.2	82.3	-9.7	84.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.0	19.8	31.6	47.6	61.6	70.7	79.1	82.4	79.2	69.5	54.7	38.9	23.5
	DBStd	22.33	5.72	7.11	8.12	7.69	6.05	4.76	3.98	4.21	5.25	6.47	7.09	6.05
	HDD50	2792	936	516	145	10	0	0	0	0	0	27	338	820
	HDD65	5457	1401	936	541	150	20	0	0	0	18	322	784	1285
	CDD50	4615	0	0	70	357	643	872	1005	906	585	174	3	0
	CDD65	1805	0	0	1	47	198	422	540	441	152	4	0	0
	CDH74	23808	0	0	42	828	2693	5435	7182	5552	1897	179	0	0
	CDH80	12484	0	0	4	261	1214	2980	4187	3038	766	34	0	0
Wind	WSAvg	5.0	4.1	4.9	5.9	6.5	6.3	5.8	5.2	4.8	4.7	4.2	4.1	3.9
Precipitation	PrecAvg	1.0	0.0	0.0	0.0	0.0	0.1	0.2	0.4	0.1	0.0	0.0	0.0	0.0
	PrecMax	4.2	0.3	0.6	0.3	0.5	0.9	1.0	3.0	1.0	0.6	0.4	0.4	0.3
	PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.7	0.1	0.1	0.1	0.1	0.2	0.3	0.5	0.2	0.1	0.1	0.1	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	57.1	78.9	92.1	96.7	102.5	104.8	102.2	95.0	84.5	66.9	44.6
		MCWB	30.8	39.6	50.0	57.1	61.7	66.8	68.8	67.9	64.3	56.6	45.8	34.0
	2%	DB	36.5	52.0	73.4	86.4	93.4	99.3	101.9	99.2	91.2	78.9	61.5	40.1
		MCWB	27.9	37.1	47.7	54.8	60.3	64.9	67.5	66.7	62.4	54.5	43.5	30.9
	5%	DB	32.9	48.5	69.4	82.6	90.4	96.6	99.3	96.6	88.0	74.6	56.9	36.6
		MCWB	25.6	35.1	46.0	53.3	59.2	64.1	66.5	66.1	61.5	52.1	41.6	28.7
	10%	DB	29.8	44.7	64.6	78.8	87.2	93.8	96.4	93.7	84.6	70.4	52.3	33.2
		MCWB	23.6	32.8	43.6	51.6	58.0	63.6	66.2	65.4	59.8	50.3	39.2	26.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.3	40.1	50.4	58.1	65.5	74.0	74.5	74.6	68.8	58.2	46.1	34.6
		MCDB	40.8	56.0	76.8	88.8	88.6	91.3	95.1	91.3	84.8	78.3	65.2	43.2
	2%	WB	28.2	37.5	48.0	55.6	62.3	69.2	71.3	71.3	65.7	55.5	43.8	31.4
		MCDB	35.9	51.0	72.4	83.9	88.2	89.4	92.7	89.3	83.4	76.3	60.5	39.2
	5%	WB	25.9	35.3	46.1	53.9	60.4	66.9	69.3	69.3	63.5	53.2	41.8	29.0
		MCDB	32.4	47.8	68.4	80.5	86.5	88.6	91.0	88.2	82.9	72.6	56.5	35.9
	10%	WB	23.7	33.0	43.9	52.1	58.9	65.4	67.9	67.6	61.5	51.0	39.4	26.9
		MCDB	29.5	44.3	64.1	77.3	84.6	87.4	90.1	87.7	80.9	68.4	51.9	32.9
Mean Daily Temperature Range	5% DB	MDBR	20.0	23.2	27.4	28.0	28.4	26.3	26.2	27.8	30.5	31.3	24.2	18.7
		MCDBR	22.9	26.4	33.2	33.6	34.2	32.9	33.4	34.2	35.6	36.0	30.2	21.4
		MCWBR	16.1	15.6	16.2	13.9	13.5	12.0	11.6	13.2	14.7	15.6	16.0	14.1
	5% WB	MCDBR	22.4	25.8	31.9	30.3	30.4	25.6	26.5	28.3	31.9	34.5	29.7	20.7
		MCWBR	15.9	15.3	15.7	13.0	13.4	11.7	11.8	13.5	14.8	14.9	15.9	13.7
Clear-Sky Solar Irradiance	taub	0.391	0.471	0.568	0.662	0.681	0.703	0.685	0.659	0.620	0.510	0.434	0.394	
	taud	1.925	1.752	1.566	1.426	1.442	1.458	1.516	1.542	1.575	1.712	1.848	1.917	
	Ebn at Noon	230	225	217	207	208	203	206	206	201	210	211	217	
	Edh at Noon	43	59	80	97	97	96	90	85	78	61	47	41	
All-Sky Solar Radiation	RadAvg	875	1171	1598	1921	2170	2198	2114	1973	1767	1420	1003	774	
	RadStd	63	60	77	115	99	127	120	109	69	41	54	55	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page